

# EE243 Advanced Computer Vision – Final Exam

Date: May 29, 2026

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1. Which of the following filters is commonly used to smooth (blur) an image and reduce high-frequency noise?
  - A. Sobel filter
  - B. Gaussian filter
  - C. Laplacian filter
  - D. Sharpening filter

2. Assume a pinhole camera projection matrix

$$P = K [R \mid t],$$

where the intrinsic matrix  $K$  is given by

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}, \quad R \in \text{SO}(3), \quad t \in \mathbb{R}^3.$$

How many degrees of freedom does  $P$  have here?

- ☐ 8
  - ☐ 9
  - ☐ 10
  - ☐ 11
3. In a ResNet residual block, the “shortcut” connection bypasses the main convolutional path and merges via which operation?
    - A. Elementwise multiplication
    - B. Elementwise addition
    - C. Channel-wise concatenation
    - D. Spatial max-pooling